

CO5430: Image Processing

Semester Project Proposal

Panorama Generation from Overlapping Photos

Project ID: P09

Group: G16

Group Members:

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1 Problem Statement

Panorama generation is a classic and fundamental problem in Computer Vision. The objective of this project is to construct a system that takes a sequence of overlapping photographs and automatically stitches them together to create a single, seamless panoramic image. This requires solving several sub-problems: detecting robust features in the images, matching those features accurately despite changes in perspective or lighting, calculating the geometric transformation (homography) between the images, and blending them together without visible seams.

2 Related Work

Feature-based image stitching has been extensively studied. Early approaches relied on corner detectors like Harris, but the introduction of the Scale-Invariant Feature Transform (SIFT) by Lowe revolutionized the field by providing features robust to scale and rotation. Later, more computationally efficient alternatives like ORB (Oriented FAST and Rotated BRIEF) were introduced, which are heavily used in real-time applications. To estimate the homography matrix in the presence of outliers (incorrect feature matches), the RANSAC (Random Sample Consensus) algorithm is the standard approach used in modern pipelines.

3 Dataset

The project will utilize a hybrid dataset approach, combining standard benchmarks with our own data, but with a strong emphasis on real-world applicability through our custom dataset:

- **Primary Custom Dataset:** Our main focus will be a self-collected dataset consisting of overlapping photographs (30-50% overlap) taken across the university campus. This will highlight the algorithm's robustness on real-world, unstructured data.
- **Supplementary Evaluation Dataset:** The *Adobe Panoramas Dataset* (<https://sourceforge.net/adobe/adobedatasets/home/Home/>) will also be used to provide a standardized benchmark for rigorous quantitative testing.

4 Methodology

4.1 Baseline Method

The baseline pipeline will rely on computationally lightweight, classical methods. It will utilize the ORB feature detector and descriptor, paired with a basic Brute-Force (BF) matcher. The homography will be estimated without advanced outlier rejection to establish a minimal working prototype.

4.2 Improved/Adapted Method

The improved system will focus on robustness and visual quality:

1. **Feature Detection:** Upgrade from ORB to SIFT to handle complex scale and rotation changes more effectively.
2. **Feature Matching:** Implement FLANN (Fast Library for Approximate Nearest Neighbors) matching combined with Lowe's Ratio Test to filter out ambiguous matches.
3. **Transformation & Optimization:** Apply RANSAC during homography estimation to discard outlier matches and calculate a highly accurate transformation matrix.

4. **Warping & Blending:** Implement image warping based on the computed homography and apply blending techniques (e.g., multi-band blending or simple feathering) to eliminate visible seams between the stitched images.

5 Evaluation Metrics

The system will be evaluated using both quantitative metrics and qualitative visual inspection across both datasets:

- **Quantitative:** Number of correct inliers found after RANSAC, and the reprojection error during homography estimation (primarily measured on the Adobe Panoramas Dataset).
- **Qualitative:** Visual alignment quality, geometric consistency, and seamless blending, with a strong focus on the final panoramas generated from our **Primary Custom Dataset**.

6 Project Timeline

Date	Milestone / Deliverable
14 Jul	M1: Proposal and Project Plan submission
28 Jul	M2: Dataset collected and ORB Baseline working
18 Aug	M3: SIFT + RANSAC prototype working; preliminary metrics gathered
25 Aug	M4: Experiment freeze; begin writing final report
01 Sep	M5: Draft report complete; repository cleaned up
07 Sep	M6: Final Submission and live demonstration

7 Risks and Mitigation

- **Risk:** Insufficient overlap in custom photos causing the feature matcher to fail.
Mitigation: Capture multiple sets of photos with varying degrees of overlap to ensure a reliable dataset.
- **Risk:** Noticeable seams or ghosting in the final panorama due to moving objects (e.g., people walking) or exposure differences.
Mitigation: Use images of static scenes for the core evaluation, and implement basic feathering blending to minimize exposure differences.

8 Task Delegation

As a 4-member group, the workload will be distributed as follows:

- **Member 1:(E/22/179)** Dataset collection, preprocessing, and Baseline (ORB) implementation.
- **Member 2:(E/22/205)** Improved feature extraction (SIFT) and FLANN matching implementation.
- **Member 3:(E/22/184)** Homography estimation, RANSAC optimization, and metric evaluation.
- **Member 4:(E/22/054)** Image warping, blending techniques, and final GitHub repository/demo pipeline integration.